

Problem-Solution Fit Canvas

Date	15 July 2026	Team ID	TM-RISINGWATERS-04
Project Name	Rising Waters: AI Powered Flood Prediction System	Maximum Marks	5 Marks

Problem-Solution Fit Canvas: Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS] • Municipal Disaster Teams: Seeking data-driven planning layouts. • Emergency First Responders: Requiring hyper-local, actionable alert scopes. • At-Risk Residents: Demanding simple, quick notifications to protect life & property.	6. CUSTOMER LIMITATIONS [CL] • Connectivity Constraints: Unstable cellular data signals during heavy storm events. • Infrastructure Budgets: Tight municipal public-funding bounds. • Tech Barriers: Diverse population with varying device models.	5. AVAILABLE SOLUTIONS [AS] • Broad Regional Weather Broadcasts: • <i>Pros:</i> Excellent geographic reach. • <i>Cons:</i> Lacks street-level inundation projections; no custom alerts.
2. PROBLEMS / PAINS + FREQUENCY [PR] • Delayed Warns: Evacuations often happen too late. • No Map Precision: Hard to see exact high-risk blocks. • Frequency: Occurs cyclically during monsoon seasons; highly devastating impacts.	9. PROBLEM ROOT / CAUSE [RC] • Traditional prediction frameworks rely on linear statistical averages. • Failure to combine active sensor arrays, topology maps, and changing weather streams dynamically in real-time.	7. BEHAVIOR + INTENSITY [BE] • Behaviors: Constantly refreshing live feeds, looking at local river markers, and manual ground monitoring. • Intensity: Very High (creates severe chaotic panic during severe weather).
3. TRIGGERS TO ACT [TR] • Abrupt severe rainfall forecasts, rising local drain/reservoir water level warnings, and historic memories of past floods.	10. YOUR SOLUTION [SL] Rising Waters: An AI-powered neural network platform fusing real-time satellite maps, topology models, and telemetry to predict street-level flood risks up to 24 hours in advance, sending instant SMS alerts to impacted zones.	8. CHANNELS OF BEHAVIOR [CH] Online: Social media trackers (X, Facebook groups), local disaster monitoring websites, municipal group chats. Offline: Emergency sirens, word-of-mouth neighbor alerts, physical line markers.
4. EMOTIONS [EM] Before: High stress, confusion, helplessness, fear. After: Relief, confidence, feeling of security, and organized proactivity.	CANVAS METRICS & FIT STRATEGY • Value Proposition: Converts panic into structured preparation by removing data gaps. • Deployment: Reaches vulnerable citizens via lightweight SMS (addressing [CL] and [CH]), requiring no smart apps or high-speed bandwidth in active crisis areas.	